IC104 Introductory Quantitative Techniques for Business and Finance

Lecture 3. Partial Differentiation and Integration

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Lecture Plan

- Partial Differentiation(Sydsaeter & Hammond Chapter 11.1-11.2)
- Integration (Sydsaeter & Hammond Chapter 9.1-9.5)

Functions of Two Variables

- **A function f of two real variables x and y with domain D** is a rule that assigns a specified number $f(x, y)$ to each point (x, y) in D.
  - $z = f(x, y)$, D is a subset of the $xy-$ plane.
- Example: A function of two variables appearing in many economic models is

$$F(x, y) = Ax^a y^b \quad \text{(A, a, and b are constants)}$$

  - Function F is called a Cobb-Douglas function.
  - F represents a production function, where x and y are input factors, and $F(x, y)$ is the number of units produced, or output.

Partial Derivatives with Two Variables

- Consider a function $z = x^3 + 2y^2$:
- Suppose y is a constant. Then z is only a function of x:

$$\frac{dz}{dx} = 3x^2$$

- Then, we keep x fixed. Then z is only a function of y:

$$\frac{dz}{dy} = 4y$$

- **Partial Derivatives**: If $z = f(x, y)$, then
  - $\frac{\partial z}{\partial x}$ denotes the partial derivative of $f(x, y)$ w.r.t x, when y is held constant;
  - $\frac{\partial z}{\partial y}$ denotes the partial derivative of $f(x, y)$ w.r.t y, when x is held constant.

Partial Derivatives: Examples

- Example 1: $f(x,y) = x^3y + x^2y^2 + x + y^2$

$$\frac{\partial f}{\partial x} = 3x^2y + 2xy^2 + 1 \quad \text{(holding } y \text{ constant)}$$
$$\frac{\partial f}{\partial y} = x^3 + 2x^2y + 2y \quad \text{(holding } x \text{ constant)}$$

- Example 2: $f(x,y) = \frac{xy}{x^2+y^2}$

$$\frac{\partial f}{\partial x} = \frac{y(x^2+y^2) - 2xxy}{(x^2+y^2)^2} = \frac{y^3 - x^2y}{(x^2+y^2)^2}$$
$$\frac{\partial f}{\partial y} = \frac{x^3 - y^2x}{(x^2+y^2)^2}$$

Higher-Order Partial Derivatives

- $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are called **first-order partial derivatives**.
- If $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are a function of (x, y), then we can define 4 functions as **second-order partial derivatives** of $f(x, y)$.

$$\frac{\partial}{\partial x}(\frac{\partial f}{\partial x}) = \frac{\partial^2 f}{\partial x^2}, \qquad \frac{\partial}{\partial x}(\frac{\partial f}{\partial y}) = \frac{\partial^2 f}{\partial x \partial y},$$
$$\frac{\partial}{\partial y}(\frac{\partial f}{\partial x}) = \frac{\partial^2 f}{\partial y \partial x}, \qquad \frac{\partial}{\partial y}(\frac{\partial f}{\partial y}) = \frac{\partial^2 f}{\partial y^2}$$

- In the previous example: $f(x, y) = x^3 y + x^2 y^2 + x + y^2$

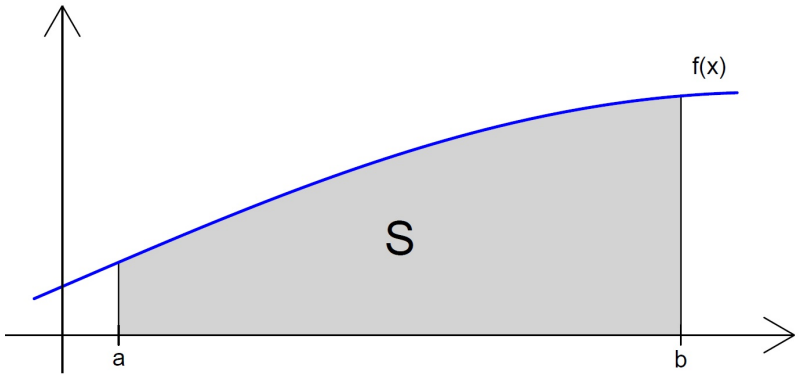
$$\frac{\partial^2 f}{\partial x^2} = 6xy + 2y^2, \quad \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = 3x^2 + 4xy, \quad \frac{\partial^2 f}{\partial y^2} = 2x^2 + 2$$

Higher-Order Partial Derivatives-Why is this important?

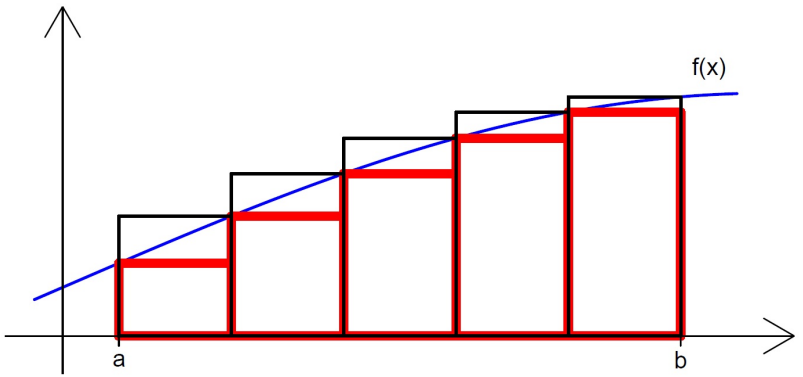
- Why is this important?
- Partial derivatives are used extensively in the pricing and hedging of derivatives.
- A bank's derivative trading desk may hold billions of Pound's worth of derivatives positions on their books. They need to insure themselves against changes in the value of the derivatives they hold, arising from changes in time, interest rate, volatility and the price of the underlying assets.
- The Black-Scholes formula is used extensively to price simple derivatives. By taking the partial derivatives of the Black-Scholes formula with respect to time, interest rate, volatility and the price of the underlying assets , we can derive exact formulas for the sensitivity of a derivatives position in each of these dimensions.
- Traders can then use these sensitivities to determine how much interest rate exposure, volatility or units of the underlying they need to buy or sell, in order to hedge their positions.

The Area Problem

- To find the area of the region S under $f(x)$ between a and b ($f(x)$ continuous on $[a, b]$):



- We create a partition P_n of the interval $[a, b]$, and use rectangles to compute upper and lower bounds for the area of S



- Consider $\lim_{n\to\infty} L(P_n, f(x))$ and $\lim_{n\to\infty} U(P_n, f(x))$.
- That is, we make each of the rectangles infinitely thin.

The Definite Integral

- If $\lim_{n\to\infty} L(P_n, f(x)) = \lim_{n\to\infty} U(P_n, f(x))$ then the definite integral of $f(x)$ from a to b is

$$\int_a^b f(x)dx = \lim_{n\to\infty} L(P_n, f(x)) = \lim_{n\to\infty} U(P_n, f(x)) = A$$

- The larger and the smaller rectangles eventually converge, to exactly match the area under the curve.
  - A is interpretable as the area under the curve S.
  - If we can calculate the area under the curve, we say that $f(x)$ is integrable

Properties

(1) $\int_a^b f(x)dx = -\int_b^a f(x)dx$

(2) $\int_a^a f(x)dx = 0$

- Graphically: no area under the graph - the rectangle has a width of zero.

(3) $\int_a^b cdx = c(b-a)$ (c: a real value constant)

- Graphically: the area under a straight line.

(4) $\int_a^b [f(x)+g(x)]dx = \int_a^b f(x)dx + \int_a^b g(x)dx$

- Graphically, this is equivalent to adding together two adjacent areas under the same curve.

Properties (continued)

$$(5)\int_a^b cf(x)dx = c\int_a^b f(x)dx \text{ (c: a real value constant).}$$

- Multiplying the function by a constant means the value of the integral is multiplied by that constant

$$(6)\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx \; a < c < b$$

- We can stick two adjacent areas under the same graph together.

$$(7)f(x) \geq 0, x \in [a,b] \Rightarrow \int_a^b f(x)dx \geq 0$$

- If a function is positive over an interval, its integral is also positive over that interval.

Properties (continued)

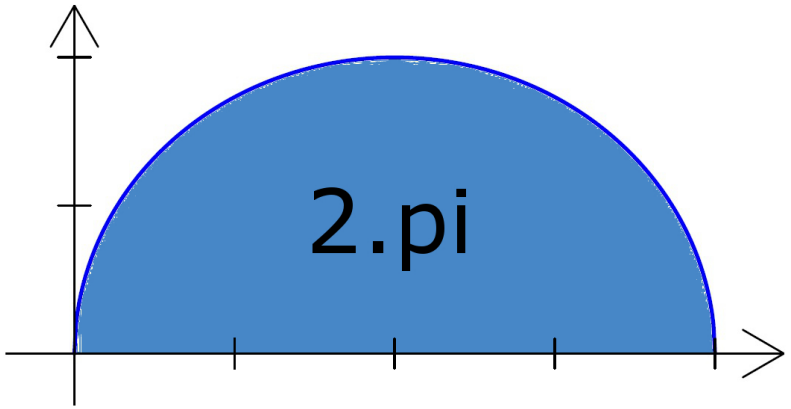
$$(8) f(x) \geq g(x), x \in [a,b] \Rightarrow \int_a^b f(x)dx \geq \int_a^b g(x)dx$$

- If a one function is greater than another over an interval, its integral is also greater than the other function's integral over that interval.

$$(9) m \leq f(x) \leq M, x \in [a,b] \Rightarrow m(b-a) \leq \int_a^b f(x) \leq M(b-a)$$

Example

- Equation of a circle: $(x - x_0)^2 + (y - y_0)^2 = r^2$



$$(x-2)^2 + (y-0)^2 = 2^2$$
$$(x^2 - 4x + 4) + y^2 = 4$$
$$y^2 = 4x - x^2$$
$$\Rightarrow y = \pm\sqrt{4x - x^2} = \pm\sqrt{4 - (x-2)^2}$$
$$\int_0^4 \sqrt{4 - (x-2)^2} = \text{Area of a Half Circle} = \frac{1}{2}\pi 2^2 = 2\pi$$

Fundamental Theorem of Calculus

- Let $f(x)$ be an integrable function. The **antiderivative** of $f(x)$ is a function $F(x)$ satisfying:

$$F(x) = \int f(x)dx \Leftrightarrow F'(x) = f(x)$$

- The antiderivative is not unique: $[F(x) + c]' = f(x)$
- We can use the **Fundamental Theorem of Calculus** to evaluate definite integrals (when a closed form of the antiderivative exists):

$$\int_a^b f(x)dx = F(x)|_a^b = F(b) - F(a)$$

- This means that if we want to find the area under the graph of $f(x)$ between a and b, we integrate it, to produce $F(x)$, and then calculate $F(b) - F(a)$.
- Definition of an Indefinite Integral:

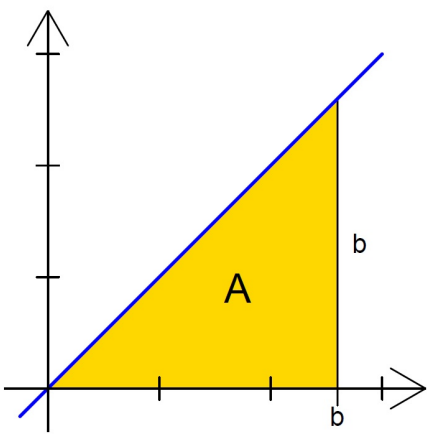
$$\int f(x)dx = F(x) + C \text{ when } F'(x) = f(x)(\text{C is a constant})$$

Example

- Suppose $F(x) = \frac{1}{2}x^2$, then $f(x) = F'(x) = x$
- The fundamental Theorem of Calculus says area under the curve $f(x) = x$ from 0 to b is:

$$\int_0^b f(x)dx = F(b) - F(0) = \frac{1}{2}b^2 - \frac{1}{2}0^2 = \frac{1}{2}b^2$$

- The area of a triangle: $A = \frac{1}{2}base \cdot height$:



- $\int_0^b f(x)dx = \int_0^b xdx = \frac{1}{2}b^2$

Evaluating Definite Integrals

- Evaluating definite integrals is, in essence, the task of finding antiderivatives.
- It's not as straight-forward as differentiation.
- The Anti-Power Rule: $\int_a^b x^n dx = \frac{1}{n+1} x^{n+1} \Big|_a^b$
- Other important integrals: the integral of a constant

$$\int_a^b c dx = cx|_a^b$$

Evaluating Definite Integrals

- Other important integrals: remember all these.

$$\int_a^b \frac{1}{x} dx = \ln x \Big|_a^b$$

$$\int_a^b e^{nx} dx = \frac{1}{n} e^{nx} \Big|_a^b$$

$$\int_a^b n^x dx = \frac{1}{\ln n} n^x \Big|_a^b, n > 0 \text{ and } n \neq 1$$

Examples

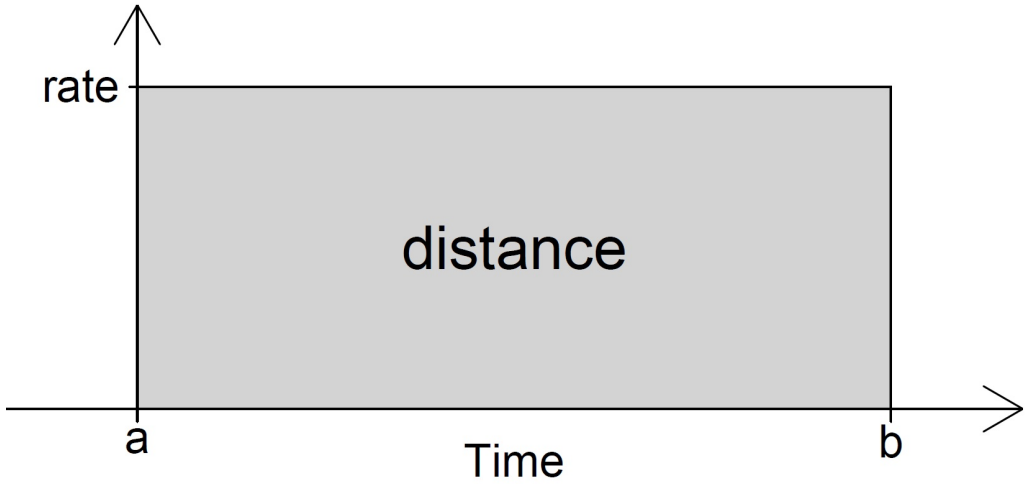
- $$\int_2^5 3x^2 + 2x + 1dx = \int_2^5 3x^2 dx + \int_2^5 2x dx + \int_2^5 1dx$$
$$= x^3\big|_2^5 + x^2\big|_2^5 + x\big|_2^5$$
$$= [5^3 - 2^3] + [5^2 - 2^2] + [5 - 2] = 141$$

- $$\int_2^5 e^{2x} dx = \frac{1}{2}e^{2x}\Big|_2^5 = \frac{1}{2}e^{10} - \frac{1}{2}e^4 = \frac{1}{2}e^4(e^6 - 1)$$

- $$\int_1^4 \frac{1}{x} dx = \ln x\big|_1^4 = \ln 4 - \ln 1$$

Applications (For interest only - not examinable)

- The Distance Formula: $distance = rate \times time$
  - Works well for distances: $100 miles = 50 mph \times 2 hours$
- Intuition for many problems. What are we really doing?



- The Job Problem:
  - If Lenny can do a job in 6 hours and Carl can do the same job in 4 hours how long does the job take if they work together?
  - Lenny can do $\frac{1}{6}$ of the job per hour, Carl can do $\frac{1}{4}$ of the job per hour
  - Together, they can do $\frac{1}{6} + \frac{1}{4}(\frac{5}{12})$ of the job per hour

The Job Problem (continued)(For interest only - not examinable)

- Here, the job is distance, so have to solve for time:

$$Time = \frac{distance}{rate} = \frac{1}{\frac{5}{12}} = \frac{12}{5} = 2 \text{ hours } 24 \text{ minutes}$$

- Solution assumes that Lenny and Carl work at a constant rate.
- Suppose instead that:
  - Lenny works at a rate $L(x) = \frac{12 - x}{54}$
  - Carl work at a rate $C(x) = \frac{8 - x}{24}$
- Expressed as definite integrals:

$$\int_0^6 L(x)dx = 1 \text{ and } \int_0^4 C(x)dx = 1$$

- The problem is to find T such that: $\int_0^T [L(x) + C(x)]dx = 1$

The Job Problem (continued) (For interest only - not examinable)

- Want to find T such that

$$\int_0^T [L(x) + C(x)]dx = 1 \Rightarrow \int_0^T [\frac{12 - x}{54} + \frac{8 - x}{24}]dx = 1$$

$$\Rightarrow 216 \cdot \int_0^T [\frac{12 - x}{54} + \frac{8 - x}{24}]dx = 216 \cdot 1$$

$$\Rightarrow \int_0^T [(48 - 4x) + (72 - 9x)]dx = 216 \Rightarrow \int_0^T [120 - 13x]dx = 216$$

$$\Rightarrow (120x - \frac{13x^2}{2})\Big|_0^T = 216 \Rightarrow 120T - \frac{13T^2}{2} - (120 \cdot 0 - \frac{13 \cdot 0^2}{2}) = 216$$

$$\Rightarrow \underline{13T^2} - 240T + 432 = 0$$

- Solve using quadratic formula: approx. 2h

Integration by Parts **(For interest only - not examinable)**

- Compute an antiderivative using the product rule backwards.
- Let $f(x)$ and $g(x)$ be continuous, integrable functions.

$$F'(x) = f(x) \quad G'(x) = g(x)$$

$$\frac{d[F(x)G(x)]}{dx} = f(x)G(x) + F(x)g(x)$$

$$F(x)G(x) = \int f(x)G(x)dx + \int F(x)g(x)dx$$

- Rearrange terms to get the integration by parts formula:

$$\int F(x)g(x)dx = F(x)G(x) - \int f(x)G(x)dx$$

- Key Idea: choose $F(x)$ and $g(x)$ so that $\int f(x)G(X)dx$ is easy to compute.

Integration by Parts: Example **(For interest only - not examinable)**

- Example 1: $\int xe^x dx$

$$F(x) = x, G(x) = e^x, f(x) = 1, g(x) = e^x$$

$$\int xe^x dx = \int F(x)g(x)dx = F(x)G(x) - \int f(x)G(x)dx$$

$$= xe^x - \int 1 \cdot e^x dx$$

$$= xe^x - e^x + C$$

- Example 2: $\int \ln(1+x)$

$$F(x) = \ln(1+x), G(x) = 1+x, f(x) = \frac{1}{1+x}, g(x) = 1$$

$$\int \ln(1+x) = \int F(x)g(x)dx = F(x)G(x) - \int f(x)G(x)dx$$

$$= \ln(1+x) \cdot (1+x) - \int (1+x) \cdot \frac{1}{1+x} dx$$

$$= (1+x)\ln(1+x) - x + C$$

Integration by Parts: Definite Integrals **(For interest only - not examinable)**

- For definite integrals:

$$\int_a^b (f(x)G(x) + F(x)g(x))dx = \int_a^b \frac{d[F(x)G(x)]}{dx}dx$$
$$= F(x)G(x)|_a^b$$
$$= F(b)G(b) - F(a)G(a)$$

- Rearrange terms to obtain:

$$\int_a^b f(x)G(x)dx = F(b)G(b) - F(a)G(a) - \int_a^b F(x)g(x)dx$$

Integration by Parts: Example (For interest only - not examinable)

- Example 3: $\int_1^3 x^2 \ln(x) dx$

$$F(x) = \ln(x), G(x) = \frac{1}{3}x^3, f(x) = \frac{1}{x}, g(x) = x^2$$

$$\int_1^3 x^2 \ln(x) dx = \int_1^3 F(x)g(x)dx = F(3)G(3) - F(1)G(1) - \int_1^3 f(x)G(x)dx$$

$$= \frac{1}{3}3^3 \ln(3) - \frac{1}{3}1^3 \ln(1) - \int_1^3 \frac{1}{x}\frac{1}{3}x^3 dx$$

$$= \frac{1}{3}27 \ln(3) - 0 - [\frac{1}{9}x^3\Big|_1^3]$$

$$= 9 \ln(3) - [\frac{1}{9}27 - \frac{1}{9}1]$$

$$= 9 \ln(3) - 3 + \frac{1}{9}$$

$$= 9 \ln(3) - \frac{26}{9}$$

Integration by Substitution **(For interest only - not examinable)**

- Integration by parts = product rule backward
- Integration by substitution = chain rule backward
- Let $F(x) = \int f(x)dx$ and $x = g(u)$. By the chain rule:

$$\frac{dF[g(u)]}{du} = F'(g(u))g'(u) = f(g(u))g'(u)$$

- Integrate both sides with respect to u

$$F(g(u)) = \int f(g(u))g'(u)du$$

- **Integration by substitution**: Let $f(x)$ be integrable and $g(x)$ invertible and continuously differentiable. The substitution $x = g(u)$ changes the integration variable from x to u as follows:

$$\int f(x)dx = \int f(g(u))g'(u)du$$

Integration by Substitution: Examples **(For interest only - not examinable)**

- Example 1: $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}$

$$\text{Let } u = \sqrt{x} \Rightarrow x = u^2 \text{ and } dx = 2u \cdot du$$

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} = \int \frac{e^u}{u} 2u du$$

$$= 2\int e^u du = 2e^u + C$$

$$= 2e^{\sqrt{x}} + C \quad (x > 0)$$

- Example 2: $\int \frac{e^x + e^{-x}}{e^x - e^{-x}} dx$

$$\text{Let } u = e^x - e^{-x} \Rightarrow du = (e^x + e^{-x})dx$$

$$\int \frac{e^x + e^{-x}}{e^x - e^{-x}} dx = \int \frac{e^x + e^{-x}}{e^x - e^{-x} dx}$$

$$= \int \frac{1}{u} du = \ln u + C$$

$$= \ln(e^x - e^{-x}) + C$$

Integration by Substitution: Definite Integrals **(For interest only - not examinable)**

- Using the Fundamental Theorem of Calculus:

$$\int_{x=a}^{x=b} f(x)dx = \int_{u=g^{-1}(a)}^{u=g^{-1}(b)} f(g(u))g'(u)du$$

$$= \int_{u=g^{-1}(a)}^{u=g^{-1}(b)} \frac{dF(g(u))}{du} du$$

$$= F(g(u))|_{u=g^{-1}(a)}^{u=g^{-1}(b)}$$

$$= F(g^{-1}(g(b))) - F(g^{-1}(g(a)))$$

$$= F(b) - F(a)$$

- Definite integral by substitution rule:

$$\int_{x=a}^{x=b} f(x)dx = \int_{u=g^{-1}(a)}^{u=g^{-1}(b)} f(g(u))g'(u)du$$

Integration by Substitution: Example (For interest only -not examinable)

- Example 3: $\int_{-1}^{0} x^2(x^3-1)^4 dx$

Let $u = x^3 - 1 \Rightarrow du = 3x^2 dx$

$$\int_{-1}^{0} x^2(x^3-1)^4 dx = \frac{1}{3}\int_{-1}^{0}(x^3-1)^4 3x^2 dx$$

$$= \frac{1}{3}\int_{u=-2}^{u=-1} u^4 du$$

$$= \frac{1}{3}\cdot\frac{1}{5}u^5\Big|_{u=-2}^{u=-1}$$

$$= \frac{1}{15}(-1)^5 - \frac{1}{15}(-2)^5$$

$$= \frac{31}{15}$$